Evaluating the 2023 unemployment insurance reform in France

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Overview

1. Labor Market Frictions and Institutional Adaptation: A Contextualization of the 2023 French Unemployment Insurance Reform
2. Research Question
3. Why Search-and-Matching Models?
4. Literature review
5. Model 1
6. Model 1.5
7. Model 2
8. Appendix

What Were We Asked to Do?

Unédic — Direction des Études et Analyses (DEA)

Current evaluations of UI reforms are **micro-focused** and concentrate on directly affected populations

They do **not** account for general equilibrium effects or macroeconomic implications

The ask

Explore macro tools for UI reform evaluation and produce first applications
to inform how DEA could integrate such analyses

Our deliverables

Research paper

Full S&M model
calibrated to France
with mathematical appendix

Simulation code

Replicable pipeline
on GitHub
ready to simulate any UI reform

This presentation

Accessible walkthrough
of the model,
calibration and key results

A first step toward macro evaluation at Unédic

The framework can be extended to any countercyclical UI reform, calibrated to any time period, and adapted to new policy parameters

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Part I

Labor Market Frictions and Institutional Adaptation: A Contextualization of the 2023 French Unemployment Insurance Reform

Beveridge Curve Shift · Moral Hazard as a Matching Friction· LMS Framework · Reform Rationale

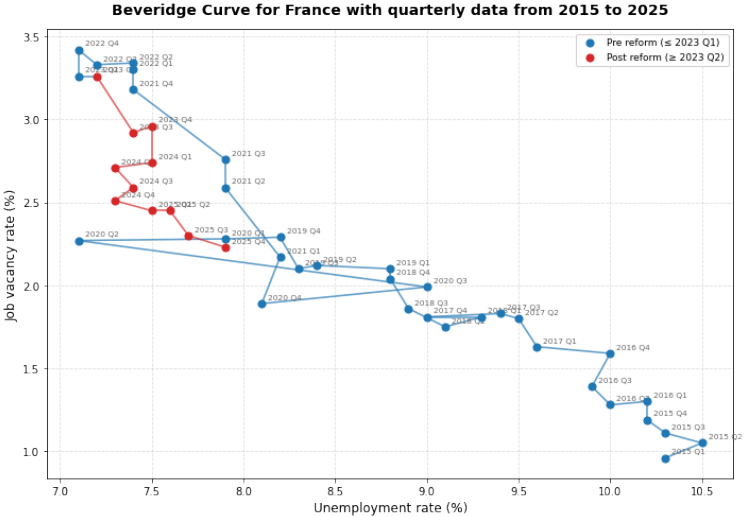
Context: The French Beveridge Curve Shift

Key facts

- Unemployment stuck at $\sim 7\%$ despite strong labor demand
- Widespread recruitment difficulties: up to 81% in construction (INSEE, 2022)

Interpretation

- Signals declining matching efficiency



A Behavioral Interpretation of the Beveridge Curve

Moral hazard as a matching friction

The Trade-off (Baily 1978; Chetty 2006)

UI benefits smooth consumption — but distort search behavior:

- ↓ search intensity
- ↑ reservation wages

$$\frac{U'(c_u) - U'(c_e)}{U'(c_e)} = \frac{\varepsilon}{1-p}$$

Insurance gain = Efficiency cost

Evidence from France

Search effort

Applications ↑ **50%** near benefit exhaustion

Marinescu & Skandalis (2021)

Unemployment duration

−10% benefit duration ⇒ −3% spell length

Le Barbanchon et al. (2019)

When Does Moral Hazard Matter? The LMS Framework

Slack labour market

Jobs are **rationed**

hiring determined by labour demand

Individual search ↑

⇒ **displacement effect**

(takes a job from another worker)

Moral hazard $\not\Rightarrow$ aggregate unemployment

Tight labour market

Binding constraint = **matching efficiency**

vacancies abundant

Individual search ↑

⇒ **direct aggregate hiring**

(no displacement)

Moral hazard ⇒ aggregate unemployment

Landais, Michaillat & Saez (2018) — augmented Baily-Chetty

$$R = \underbrace{\frac{u'(c_u) - u'(c_e)}{u'(c_e)} \cdot \frac{1-p}{\varepsilon}}_{\text{Baily–Chetty term}} + \underbrace{\Phi(\theta) \cdot \left(\frac{\varepsilon^{\text{mac}}}{\varepsilon^{\text{mic}}} - 1 \right)}_{\text{elasticity wedge}}$$

The 2023 reform as a direct application of LMS state-dependent logic

Post-2015: Beveridge curve shifts ⇒ macro-elasticity → micro-elasticity ⇒ wedge closes ⇒ UI generosity should fall
LMS: adjust **replacement rate** France 2023: adjust **benefit duration** same underlying logic

Why Adjust Duration Rather Than the Replacement Rate?

Three possible UI levers

Replacement rate
Historically resisted
Abolished in 2001 (degressivity)
Welfare loss for low-income

Benefit duration ✓
2023 reform lever
Targets timing of incentives
Preserves early-spell insurance

Eligibility conditions
Primary lever historically
2019: threshold raised
from 4 to 6 months

Why duration works
Preserves income support early in the spell
when liquidity needs are strongest
Increases incentives as spell **lengthens**
End-of-benefit acts as a **psychological anchor**

Bonnet et al. (2024)
36% of income loss offset
immediately by spending cuts
↑ 46% after 6 months
MPC higher for low-income

Two macroeconomic triggers for the 25% duration cut

Structural threshold
Unemployment rate **below 9%**

Above: job rationing regime
⇒ full duration restored

Momentum trigger
+0.8 pp rise within a single quarter

Detects sudden deterioration
⇒ immediate full duration restoration

Retrospective validation

Both triggers would have accurately identified the **2009 GFC** and **2020 COVID** shocks — see next slide

Retrospective Validation of the Two Triggers

2009 — GFC

Momentum trigger (+0.8pp) activates full duration

Structural rule (9%) maintains full benefits until 2018

2020 — COVID

Sudden spike triggers immediate full restoration

Automatic response without discretion

The rule tightens when the market is tight, relaxes when it is not

Exactly the state-dependent logic of Landais, Michaillat & Saez (2018)

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Part II

Research Question

Policy Objectives · Parliamentary Debate · Analytical Framework

Policy Objectives of the 2023 Reform

Minister Olivier Dussopt, National Assembly, September 13 2022

"It is dominated by one objective: full employment [...]
when the economy is good, the rules must be more incentive-based."

Three stated objectives

1. Full Employment

Reduce unemployment to a residual frictional level (target: below 5% ILO)

"Full employment is certainly an economic objective, but also a political and social one."

2. Work Incentives

Reinforce incentives to return to work during favourable economic conditions

"Too protective [during booms], it loses its incentive character."

3. Recruitment

Alleviate widespread hiring difficulties reported by firms across all sectors

"The number of unfilled vacancies per 1,000 seekers rose from 50 to 170."

Common causal claim

Changing UI rules $\Rightarrow$ accelerate transitions into employment $\Rightarrow$ ease firms' hiring constraints

A Parliamentary Counterpoint

Deputy Olivier Serva (LIOT), National Assembly 2022

"This text claims to address recruitment difficulties,
in reality deals with unemployment insurance —
as if the link between the two were obvious and automatic."

Three objections that motivate our analysis

Full employment
Has enough time elapsed since the previous reform to justify further tightening?

Work incentives
Does the existing system not already encourage return to employment?

Recruitment difficulties
Only 40% of job seekers receive benefits — tensions are multifactorial

Effects of UI reform cannot be inferred from intuition alone
They must be analysed within a framework identifying the conditions under which benefit duration affects search effort, vacancy creation, and hiring outcomes

Research Question

Central Research Question

To what extent is the design of the 2023 French unemployment insurance reform consistent with the policy objectives articulated by the government?

We focus on three specific dimensions

1. Full employment
Can a reduction in benefit duration contribute to increasing employment?

2. Incentives
Is the contracyclical modulation sufficiently calibrated to strengthen job-search effort?

3. Recruitment tensions
Can the reform alleviate the recruitment difficulties reported by firms?

Our approach

Search-and-matching model with endogenous search effort

Calibrated to the French labour market (H2 2022)

Steady-state comparison: pre-reform vs. post-reform

Reform $= \lambda \uparrow \Rightarrow D \downarrow 25\%$

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Part III

Why Search-and-Matching Models?

Walrasian Limits · Matching Function · State-Dependent Policy

Why Search-and-Matching Models?

A framework suited to the 2023 reform

The Walrasian failure

In a spot market with flexible wages, unemployment reflects only rigidities — UI affects *labor supply only*, firms are passive.

$\Rightarrow$ Cannot explain coexisting unemployment & vacancies in France pre-2023.

Three reasons S&M fits

- Explains the **Beveridge curve shift** as a matching efficiency outcome
- Links **micro behavior** (reservation wages, search effort) to macro equilibrium
- Accommodates **state-dependent policy** — essential for the reform's trigger logic

Why Search-and-Matching Models?

Walrasian models cannot account for the coexistence of unemployment and vacancies

In a spot market, wages clear instantly — UI only affects labour supply, with no implications for vacancy creation or hiring

Four S&M ingredients directly relevant to the 2023 reform

Matching function $m(u, v)$

Hiring requires coordination —
UI affects search intensity,
altering the rate of match formation

Labour market tightness $\theta = v/u$

State variable of the reform:
benefit duration is conditioned
on unemployment — a proxy for θ

Endogenous vacancy creation

Firms respond strategically to UI:
shorter benefits $\Rightarrow$ lower wages
$\Rightarrow$ more vacancies posted

Nash bargaining & search effort

UI shapes outside option V_l
$\Rightarrow$ wages and search effort
respond endogenously

S&M models allow policy effects to vary across macroeconomic regimes

Essential for a reform explicitly designed to be **state-dependent** — tightening when θ is high, relaxing when θ is low

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Part IV

Literature Review

Search Incentives · Wage Setting · Vacancy Creation · Macro UI

Presenting the structure of the literature review

As a whole, the macroeconomic literature is much scarcer than the micro literature. We draw on two complementary strands depicting core mechanisms of our model.

Strand 1: Search incentives & job finding behaviors

How does UI generosity affect **individual search behavior**?

Key mechanism: *Endogenous search effort* s_i and s_u and reservation wages as functions of benefit duration

Key papers: Mortensen (1977), Baily (1978), Chetty (2008), Marinescu & Skandalis (2021), Le Barbanchon et al. (2019)

Strand 2: Wage setting, vacancy posting & equilibrium unemployment

How does UI affect **firm-side decisions** and aggregate unemployment?

Key mechanism: *Nash bargaining* linking outside options V_l to wages w and vacancy creation v under free entry

Key papers: Fredriksson & Holmlund (1998), Landais, Michaillat & Saez (2018), Brébion et al. (2025), Rioux & Ortega (2010)

Fredriksson & Holmlund (1998), Optimal UI in search equilirbium

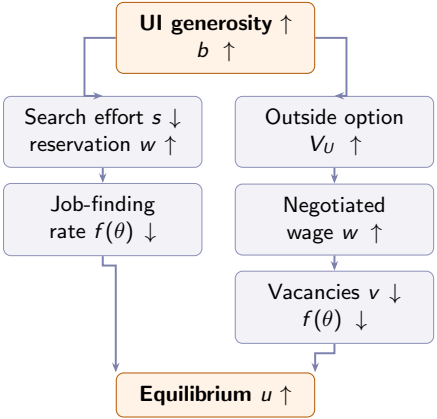
Core argument: standard search models focus on how UI affects workers. Fredriksson & Holmlund show that UI also operates through *wages and vacancy creation*. Proving that firm channels are quantitatively important.

The mechanism:

- UI raises V_U (value of unemployment) & workers outside option
- Nash bargaining $\Rightarrow$ higher negotiated wage w
- Higher w lowers firm surplus $J - V_v$
- Under free entry, fewer vacancies posted
- Fewer vacancies $\Rightarrow$ lower $f(\theta)$ $\Rightarrow$ higher u

The wage channel reduces vacancy creation, which amplifies the moral hazard channel. Unlike other contemporary papers, they characterize the optimal benefit as limited and declining over the unemployment spell.

Graphical representation of the dual impact channels of UI schemes:



Both channels increase equilibrium unemployment

Literature synthesis, search incentives contributions (1/2)

Paper	Mechanism & research question	Model & key dynamics
Mortensen (1977)	*Mechanism:* UI raises the value of unemployment, affecting search effort and job acceptance. The Mortensen-Pissarides model constitutes a real reference for academic research modeling unemployment insurance.	Partial equilibrium job search model with endogenous search effort. *Key result:* More generous UI schemes can increase unemployment duration by reducing search intensity.
Baily (1978); Chetty (2008)	*Mechanism:* Highlights the trade-off between consumption smoothing and search distortions. Baily introduced the framework, Chetty (2008) is the modern sufficient statistics restatement (widely cited in optimal UI design papers)	Sufficient-statistics approach: optimal UI depends on consumption drop & unemployment duration elasticity from a welfare perspective

These papers provide the benchmarks of micro foundations for modeling endogenous in the academic field, through search effort and the insurance to search incentive trade-off.

Literature synthesis, search incentives contributions (2/2)

Paper	Mechanism & research question	Model & key dynamics
Marinescu & Skandalis (2019)	Benefit duration affects timing of search effort. *Mechanism:* Introduces within spell dynamics of search behavior and documents strong timing effects of UI duration through a pre exhaustion surge in applications.	Empirical search model with finite benefit duration and eligibility conditions, calibrated on French system. Search effort rises sharply near exhaustion; adjustment occurs mainly through timing. They quantify a 50% spike in applications right before benefits exhaustion.
Le Barbanchon et al. (2019)	*Mechanism:* Effect of potential benefit duration on reservation wages, job finding and job selectivity. Shows that reservation wages respond weakly to UI duration from an empirical analysis relying on the 2009 French UI reform, introducing altering entitlement duration for elegible workers.	*Key result:* Weak response of reservation wages; duration mainly affects job-finding behavior, documenting a 3% reduction in spell length per 10% cut in duration from a regression discontinuity on French data. Justifies focusing on search effort rather than reservation wage as an adjustment mechanism in certain models.

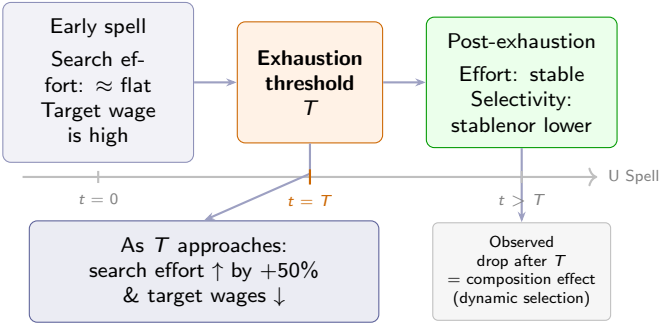
Empirical evidence suggests that UI primarily affects search timing rather than wage setting behavior which aligns with our model design.

Marinescu & Skandalis (2019), UI and job search behavior

Core argument: UI shapes job search effort dynamically through effort and selectivity. Quantify strong anticipation of benefit exhaustion (French calibrated study).

Key results:

- Quantify a +50% increase in search intensity as exhaustion approaches
- Target wages decline progressively before exhaustion & post exhaustion behavior remain constant, driven by dynamic selection



Relevance for the 2023 reform

Shortening benefit duration T shifts incentives earlier in the spell: it raises search effort and lowers target wages before exhaustion.

Literature synthesis, wages and vacancy creation publications (1/2)

Paper	**Mechanism & research question**	**Model & key dynamics**
Fredriksson & Holmlund (1998)	*Mechanism:* UI affects unemployment through simultaneous channels: wage bargaining and vacancy posting in a single equilibrium framework.	Motivates our model architecture with wage bargaining and vacancy creation. *Key result:* UI raises wages, reduces vacancy posting, and increases unemployment.
Landais, Michaillat & Saez (2018)	*Mechanism:* Labor market tightness modifies the welfare cost of UI.	GE matching model with vacancy creation. *Key result:* Optimal UI is state-dependent across the business cycle.

The above mentioned papers show that UI affects aggregate outcomes through wages, vacancy posting, and at equilibrium, labor market tightness.

Landais, Michaillat & Saez (2018) Macroeconomic approach to optimal UI

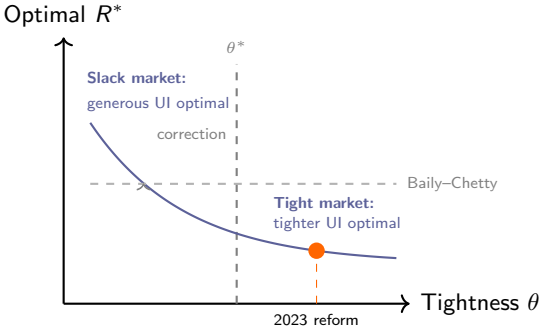
Intuition: the Baily-Chetty formula is incomplete when labor market is not at efficient tightness θ^*. The aggregate welfare or social cost of moral hazard depends on where the economy sits on the Beveridge curve.

Augmented formula:

$$R^* = \underbrace{\frac{u'(c_u) - u'(c_e)}{u'(c_e)} \cdot \frac{1-p}{\varepsilon}}_{\text{Baily–Chetty term}} + \underbrace{\text{tightness correction}}_{\text{macro correction term}}$$

The correction term captures how UI affects θ through vacancy creation. When $\theta < \theta^*$, more generous UI may raise welfare despite moral hazard (=2023 reform transposes this logic via counter cyclical modulation).

Optimal UI across the business cycle:



In tight markets, the additional counter cyclical tightness correction term reduces the optimal replacement rate below the Baily-Chetty benchmark.

Literature synthesis, wages and vacancy creation publications (2/2)

Paper	Mechanism & research question	Model & key dynamics
Rioux & Ortega (2010)	*Model:* Search model with non automatic endogenous eligibility and re-entitlement effects that together link employment to future UI access	*Key result:* Reduces wage pressure and unemployment through forward-looking incentives. The major contribution is the study of re-entitlement effects: the fact that taking a job conditions future access UI access and changes the value of employment.
Brébion et al. (2025)	*Mechanism:* UI and more specifically eligibility thresholds may affects contract design offered by firms from an ex-ante response of firms.	Empirical analysis of legislative changes in eligibility thresholds, exploiting an UI reform. *Key result:* Firms adjust contract durations, prior to match formation (ex-ante) in response to UI eligibility rules as they internalize workers preferences. Hence affecting shaping vacancy distribution.

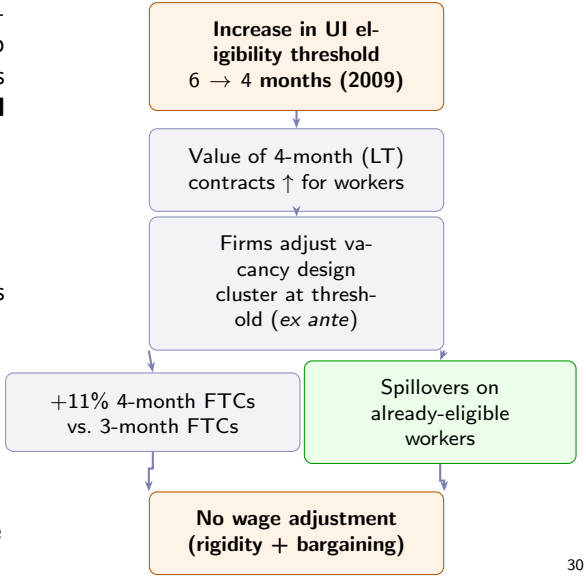
Such extensions motivate key features of our model and present features such as eligibility conditions and finite entitlements that are consistent with the French design.

Brébion, Briole & Khoury (2025), UI eligibility & employment duration

Core argument from a French empirical evaluation: UI eligibility conditions not only shape job search behaviors but also firms' hiring decisions through an **ex-ante vacancy design channel operating before the match**.

Key results:

- An 11% increase in longer fixed term contracts offered vs shorter ones post-reform to meet workers' expectations
- Increased vacancy posting from firm channel in response to UI modulation (relatively more 4-month vacancies *ex ante*)
- Document significant spillover effects on already eligible workers: macro effects are found to exceed micro estimates



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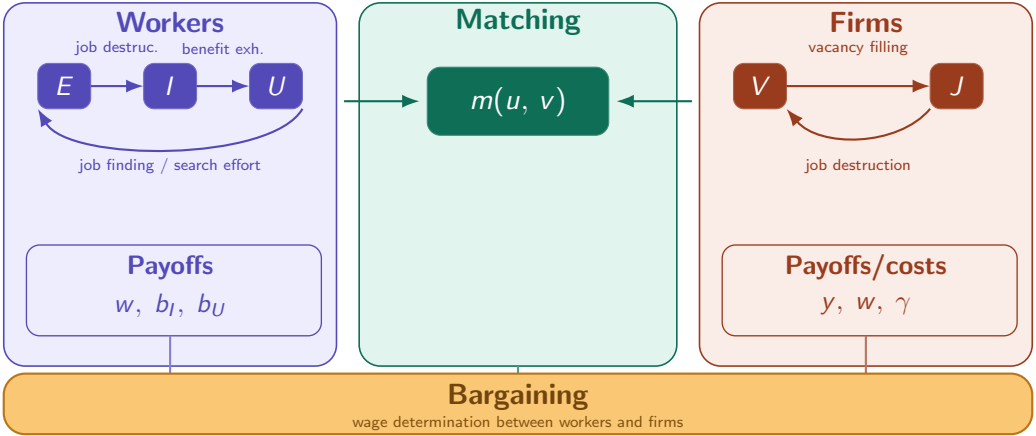
Part V

Model 1
Baseline Search-and-Matching

Setup · Value Functions · Calibration · Results

How We Model the Labour Market



How S&M Ingredients Enter in This Model

v = vacancies posted by firms

Labour market tightness

$$\theta \equiv \frac{v}{u}$$

u = effective job searchers $s_I u_I + s_U u_U$

Job finding rate $f(\theta)$

Prob. a worker finds a job

$$f(\theta) = \frac{m(u,v)}{u} = m(1,\theta)$$

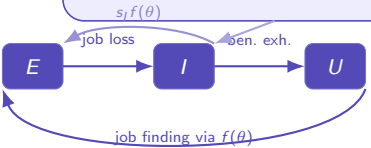
$f'(\theta) > 0$: tighter $\Rightarrow$ easier to find a job

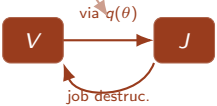
Vacancy filling rate $q(\theta)$

Prob. a firm fills a vacancy

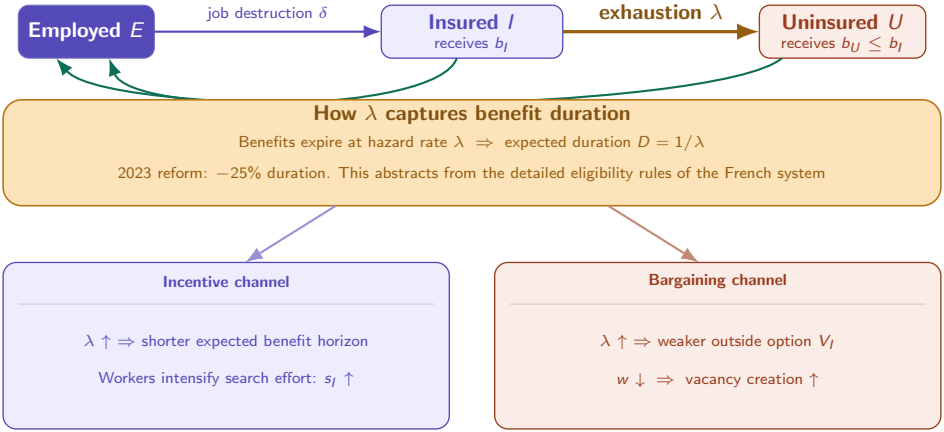
$$q(\theta) = \frac{m(u,v)}{v} = m(1/\theta, 1)$$

$q'(\theta) < 0$: tighter $\Rightarrow$ harder to fill





A Focus on How We Model Unemployment Insurance



Formalization : How Workers and Firms Behave in our Model

Workers

Employment

$$rV_E = w + \delta(V_I - V_E)$$

Insured unemployment

$$rV_I = b_I - \frac{1}{2}c_0 s_I^2$$
$$+ s_I f(\theta)(V_E - V_I) + \lambda(V_U - V_I)$$

Uninsured unemployment

$$rV_U = b_U - \frac{1}{2}c_0 s_U^2$$
$$+ s_U f(\theta)(V_E - V_U)$$

Firms

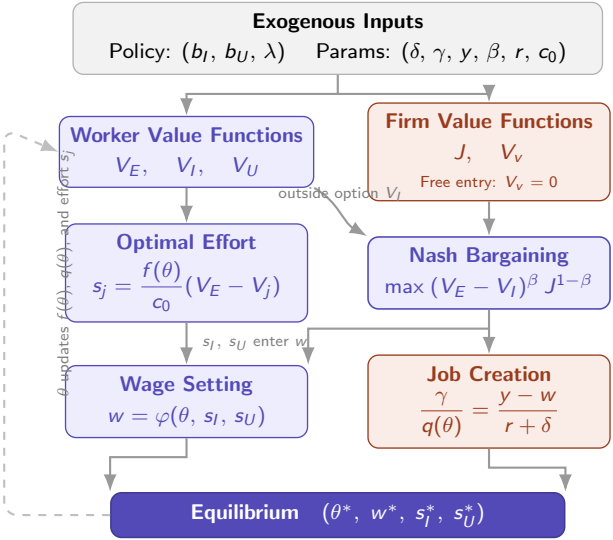
Filled job

$$rJ = y - w + \delta(V_v - J)$$

Open vacancy

$$rV_v = -\gamma + q(\theta)(J - V_v)$$

How the Model Solves



In Theory, This is Pinned Down by the Following Fystem of Equations

(i) Job creation

$$\frac{\gamma}{q(\theta)} = \frac{y - w}{r + \delta}$$

(ii) Wage

$$w = b_I - \tfrac{1}{2} c_0 s_I^2 + \left(r + \delta + s_I f(\theta) + \lambda\right) \frac{\beta}{1-\beta} \frac{\gamma}{q(\theta)} - \lambda \frac{c_0}{f(\theta)} s_U$$

(iii) Insured search effort

$$s_I = \frac{f(\theta)}{c_0} \frac{\beta}{1-\beta} \frac{\gamma}{q(\theta)}$$

(iv) Uninsured search effort

$$\tfrac{1}{2} c_0 s_U^2 + \frac{c_0 (r + \lambda)}{f(\theta)} s_U + \left[(b_U - b_I) - (r + \lambda) \frac{\beta}{1-\beta} \frac{\gamma}{q(\theta)} - \frac{f(\theta)^2}{2c_0} \left(\frac{\beta}{1-\beta} \frac{\gamma}{q(\theta)} \right)^2 \right] = 0$$

Calibration and Simulation Strategy

Goal

Compare two **steady-state equilibria**: pre-reform vs. post-reform

If the French labour market had always operated under reformed rules, what would the long-run equilibrium look like?

Step 1 — Calibration to H2 2022

Fix r, β, η, y from the literature

Estimate $(\delta, \lambda, \gamma, m_0, b_I, b_U)$ by minimising distance to French data:

unemployment rate • vacancy rate • long-term unemployment • UI duration

Step 2 — Reform Simulation

Increase λ by 25%: $\lambda_{\text{reform}} = \lambda_{\text{base}} / 0.75 \quad \Rightarrow \quad D_{\text{reform}} \approx 3.98$ quarters

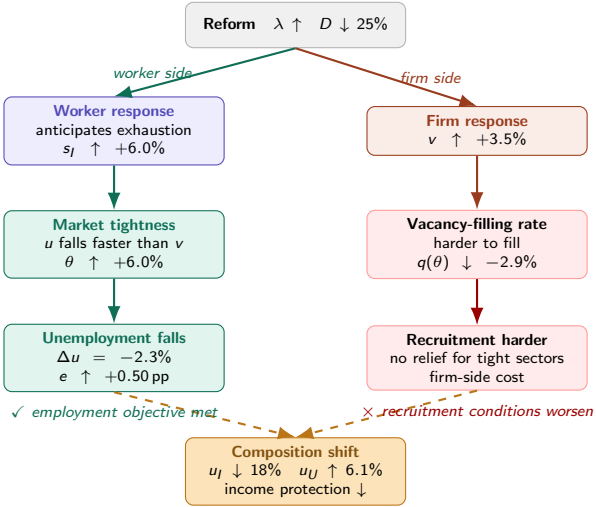
All other parameters held fixed

Re-solve equilibrium system $\Rightarrow$ new $(\theta^*, w^*, s_I^*, s_U^*)$

What we read off

Changes in θ, w, s_I, s_U, insured vs. uninsured unemployment

Results


Reform $\lambda \uparrow$ $D \downarrow 25\%$
worker side
firm side
Worker response
anticipates exhaustion
$s_I \uparrow +6.0\%$
Firm response
$v \uparrow +3.5\%$
Market tightness
u falls faster than v
$\theta \uparrow +6.0\%$
Vacancy-filling rate
harder to fill
$q(\theta) \downarrow -2.9\%$
Unemployment falls
$\Delta u = -2.3\%$
$e \uparrow +0.50$ pp
Recruitment harder
no relief for tight sectors
firm-side cost
✓ employment objective met
✗ recruitment conditions worsen
Composition shift
$u_I \downarrow 18\%$ $u_U \uparrow 6.1\%$
income protection $\downarrow$

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Part VI

Model 1.5
Wage Rigidities Extension

SMIC · Wage Rigidities · Bargaining Channel · Comparison

Wage Rigidities: Motivation & Key Takeaways

France: one of the most binding minimum wage regimes in the OECD

The *Salaire Minimum Interprofessionnel de Croissance* (SMIC) sets a wage floor across all sectors.

We therefore extend the baseline model by introducing a minimum wage w_0 below which the negotiated wage cannot fall.

When $w_0 \geq w^*$: the SMIC binds.

Consequence for the 2023 reform

The reform reduces unemployment through the incentive channel only

The bargaining channel is shut down by the wage floor

The employment effect is weaker than without the wage floor (-0.33pp vs -0.50pp)

Two Regimes

Modification: impose a wage floor w_0

Workers and firms can no longer bargain freely below w_0

Case 1: $w_0 < w^*$

Non-binding constraint

Nash bargaining delivers w^*

θ and w jointly determined

as in Model 1

Equilibrium of Model 1 unchanged

No further analysis needed

Case 2: $w_0 \geq w^*$

Binding constraint

$w = w_0$ fixed exogenously

Simultaneity between θ and w breaks down

$$w_0 \geq w^* \;\Rightarrow\; \theta_0 \leq \theta^*$$

New sequential system

Structural consequence

θ_0 is pinned by the wage floor w_0, it cannot be bargained away

(s_I, s_U) adjust subsequently, taking θ_0 as given

Formalization: The New Equilibrium System

(i) Job creation — pins down θ_0 directly

$$\frac{\gamma}{q(\theta_0)} = \frac{y - w_0}{r + \delta} \qquad w_0 \text{ replaces } w^*;\ \theta_0 \leq \theta^* \text{ since } w_0 \geq w^*$$

(ii) Insured effort — derived without Nash

$$(r + \delta + \lambda)\,\frac{c_0\, s_I}{f(\theta_0)} = (w_0 - b_I) - \tfrac{1}{2} c_0 s_I^2 + \lambda\,\frac{c_0\, s_U}{f(\theta_0)}$$

(iii) Uninsured effort — derived without Nash

$$(r + \lambda)\,\frac{c_0(s_U - s_I)}{f(\theta_0)} = (b_I - b_U) + \tfrac{1}{2} c_0\left(s_I^2 - s_U^2\right)$$

How the Model Solves Under Wage Rigidity

Exogenous Inputs

Policy: $(b_I,\ b_U,\ \lambda,\ w_0)$ Params: $(\delta,\ \gamma,\ y,\ r,\ c_0)$

Step 1 — Job Creation

$$\frac{\gamma}{q(\theta_0)} = \frac{y - w_0}{r + \delta}$$

$$\longrightarrow\ \theta_0$$

Step 2 — Optimal Effort

$$s_j = \frac{f(\theta_0)}{c_0}(V_E - V_j), \quad j \in \{I, U\}$$

$$\longrightarrow\ s_I,\ s_U$$

Equilibrium $(\theta_0,\ s_I^*,\ s_U^*)$ $w = w_0$ *(exogenous)*

Calibration and Simulation Strategy

Same calibration as Model 1 — same parameters, same targets

All parameters $(\delta, \lambda, \gamma, m_0, b_I, b_U, c_0)$ held at calibrated values
The only addition: $w_0 = 0.9$, which exceeds the Nash wage $w^* = 0.880$

Reform Simulation — identical to Model 1

$\lambda_{\text{reform}} = \lambda_{\text{base}} / 0.75 \quad \Rightarrow \quad D_{\text{reform}} \approx 3.98$ quarters

All other parameters held fixed

Re-solve system $\Rightarrow$ new $(\theta_0,\ s_I^*,\ s_U^*)$

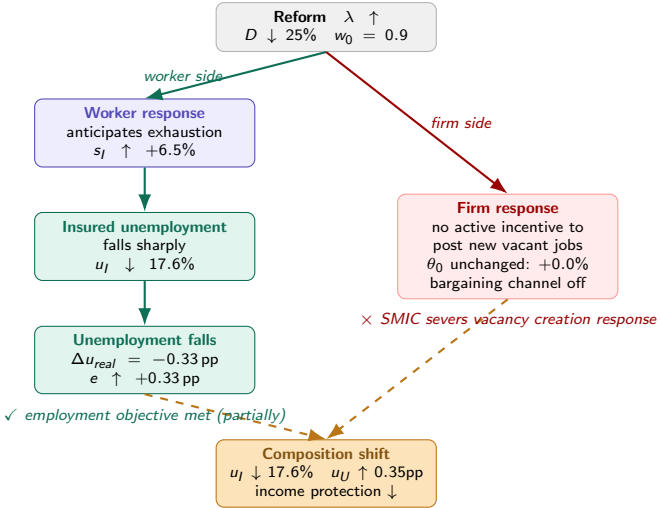
What we read off

Changes in θ_0, s_I, s_U, insured vs. uninsured unemployment

Central question

Does the SMIC attenuate the reform's employment effect
by severing the link between benefit duration and vacancy creation?

Results Under Wage Rigidity



Comparing Model 1 and Model 1.5

	Model 1 *(Nash bargaining)*	**Model 1.5** *(Minimum wage $w_0 = 0.9$)*	**Difference**
$\Delta\theta$	$+6.0\%$	0.0%	-6.0pp
Δs_I	$+6.0\%$	$+6.5\%$	$+0.5$pp
Δs_U	-0.0%	-0.2%	-0.2pp
Δ total unemployment	-0.50pp	-0.33pp	$+0.17$pp
Bargaining channel	Active	Shut down	—
Incentive channel	Active	Active	—

Key takeaway

The SMIC does not prevent the reform from reducing unemployment,
but severs the link between benefit duration and vacancy creation,
reducing the total employment gain by one third (-0.50pp $\rightarrow$ -0.33pp)

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Part VII

Model 2
Endogenous Job Destruction & Business Cycles

Idiosyncratic Shocks · Markov States · Match Surplus · Beveridge Dynamics

Toward More Realistic Model : Endogenous Job Destruction & Business Cycles

Key Shift: From exogenous shocks to a choice to terminate + business cycles dynamics.

- **Endogenous Threshold (R_i):** allowed by differentiation in job productivity $p_i + \sigma y$
  - Matches are destroyed when y falls below a state-dependent reservation level R_i.
- **Macroeconomic States:** 3-state Markov process (Boom, Intermediate, Recession with $p_1 > p_2 > p_3$) with transition matrix Ω_{ij}.
- **No Endogenous Search Effort:** no incentive channel.

Two Sources of Job Destruction

All new jobs are created at maximum productivity. Over time, the value of a job can change due to:

1. Job-Specific Shock ($y \to y'$)

- At rate μ
- Firm decides whether continue producing or destroy the job
- Destroy when $J(y') \leq 0 \iff$ y'$\leq R_i$

2. Macroeconomic Shock ($p_i \to p_j$)

- At transition probabilities Ω_{ij}
- Reservation threshold R_j changes
- Job can become unviable if $y < R_j$

Formalization : How Workers and Firms Behave in our Model

Workers

EMPLOYMENT

$$rV_{e_i}(y) = w_{e_i}(y) + \mu\left[\int \max\{V_{e_i}(y'), V_{I_i}\}\, dG(y') - V_{e_i}(y)\right]$$

$$+ \sum_{j\neq i}^{n} \Omega_{ij}\left[\max\{V_{e_j}(y), V_{I_i}\} - V_{e_i}(y)\right]$$

INSURED UNEMPLOYMENT

$$rV_{I_i} = b_I + \theta_i q(\theta_i)\left(V_{e_i}(\bar{y}) - V_{I_i}\right)$$

$$+ \lambda(V_{U_i} - V_{I_i}) + \sum_{j\neq i}^{n} \Omega_{ij}(V_{I_j} - V_{I_i})$$

UNINSURED UNEMPLOYMENT

$$rV_{U_i} = b_U + \theta_i q(\theta_i)\left[V_{e_i}(\bar{y}) - V_{U_i}\right]$$

$$+ \sum_{j\neq i}^{n} \Omega_{ij}(V_{U_j} - V_{U_i})$$

Firms

FILLED JOB

$$rJ_i(y) = p_i + \sigma y - w_i(y) + \mu\left[\int \max\{J_i(y'), V_i\} dG(y')\right.$$

$$\left. - J_i(y)\right] + \sum_{j\neq i}^{n} \Omega_{ij}\left[\max\{J_j(y), V_j\} - J_i(y)\right]$$

OPEN VACANCY

$$rV_{v_i} = -\gamma + q(\theta_i)\left[J_i(\bar{y}) - V_{v_i}\right]$$

$$+ \sum_{j\neq i}^{n} \Omega_{ij}(V_{v_j} - V_{v_i})$$

Equilibrium Conditions

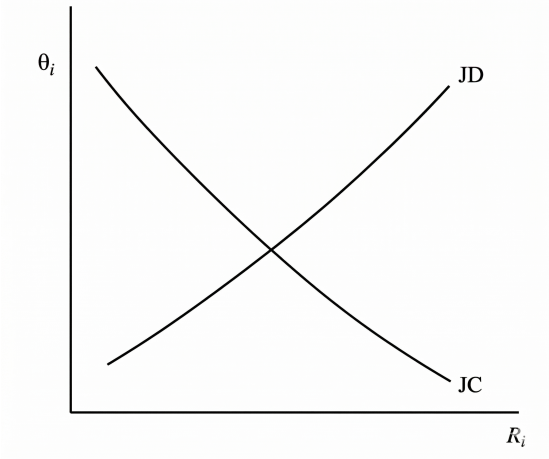


Figure: Equilibrium Job Creation and Job Destruction

Job Creation Condition

- Vacancy cost = firm's share of surplus

Job Destruction Condition

- Minimal productivity to keep a match alive = match cost

Equilibrium outcome: $(R_i, \theta_i) \Rightarrow$ unemployment u_i via Beveridge curve

New Mechanisms

Mechanism		Job Creation	Job Destruction
Macroeconomic (p_i)	**State**	Better conditions $\rightarrow$ more vacancy posting	Better conditions $\rightarrow$ lower survival threshold
Macroeconomic Value	**Option**	Anticipated cost changes $\rightarrow$ creation postponed in booms, brought forward in recessions	Expected improvements $\rightarrow$ marginal jobs preserved (less destruction in recessions, muted rise in booms)
Idiosyncratic Value	**Option**	Higher future match value $\rightarrow$ more creation	Labour hoarding $\rightarrow$ firms retain low-productivity jobs

Table: Mechanisms affecting job creation and destruction

The Beveridge Curve Dynamics

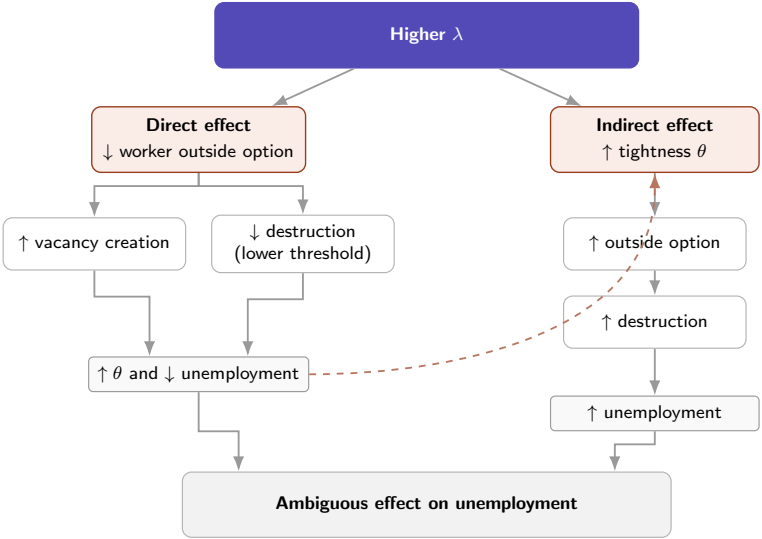
The relationship between u and v is driven by two opposing forces:

1. **Matching Effect (-):** Higher vacancies $\rightarrow$ higher θ $\rightarrow$ faster outflows from unemployment (standard downward slope).
2. **Job Destruction Channel (+):** Higher v increases the worker's outside option. This rises the reservation threshold R_i.

Implication: If the JD channel is strong, the Beveridge curve can flatten or become upward sloped, as higher turnover offsets faster hiring.

$$\boxed{u_i = \frac{\mu G(R_i)}{\mu G(R_i) + \theta_i q(\theta_i)}}$$

Effects of Increasing Benefit Expiration Rate



An Illustration : Policy Effects

Reform $\lambda \uparrow \quad D \downarrow 25\%$

Variable	**Baseline**	**Policy Change**	**Difference**
Market Tightness	0.63	0.67	+0.04 pp
Reservation Productivity	1.9	1.89	-0.01
Job Destruction	14.23%	14.21%	-0.02 pp
Job Creation	1.58	1.63	+ 0.05
Vacancy Rate	5.10%	5%	-0.1 pp
Unemployment	8.34%	8.09%	-0.25 pp
Insured Unemp.	7.43%	7%	-0.43
Uninsured Unemp.	0.89%	1.09%	+0.2

Table: Policy Experiment : Average Values Across the Business Cycle

Conclusion — How Unemployment Insurance Transmits to the Economy

Main takeaway

UI affects search, wages, vacancies and separations;
these effects are **state-dependent** over the business cycle

Four transmission channels

1. Search incentives

$D \downarrow \Rightarrow$ earlier exhaustion
$\Rightarrow s_l \uparrow$
$\Rightarrow$ faster exits

2. Wage bargaining

Less generous UI
$\Rightarrow$ weaker outside option
$\Rightarrow w \downarrow$

3. Vacancy creation

$w \downarrow \Rightarrow$ surplus $\uparrow$
$\Rightarrow v \uparrow,\ \theta \uparrow$
$\Rightarrow f(\theta) \uparrow$

4. Job destruction

UI affects match surplus,
separation risk,
and job destruction

Across the business cycle

In **tight markets**, search and vacancy effects translate more into aggregate hiring;
in **slack markets**, displacement is stronger and insurance is more valuable

Final implication for the 2023 reform

The reform follows a **state-dependent macro logic**: its effects depend on labour market tightness,
business-cycle conditions, and institutional constraints such as wage rigidities

Next Steps: From a Theoretical Toolbox to Policy Evaluation

We provided a structural toolbox linking benefit duration to the Beveridge curve, individual search incentives, job creation and job destruction decisions.

- **Empirical Calibration & Simulation:** Adapt the model with endogenous job destruction to your preferred labour market moments.
- **Model Sophistication:**
  - Merging the extensions by incorporating **endogenous search effort** and the **minimum wage** into the business-cycle framework.
  - Introducing **state dependent benefit levels and benefit expiration rate**
  - Accounting for **heterogeneity** through workers' skills or sectors
- **Transition Dynamics:** Moving beyond steady-state comparisons to model the out-of-equilibrium path.

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Match Surplus: the Engine of Labour Market Dynamics

Match Surplus $S_{e_i}(y)$: value of a worker–firm match relative to outside options; it governs both job creation and destruction.

$$\underbrace{(r+\mu+\sum_{j\neq i}\Omega_{ij})S_{e_i}(y)}_{\text{Match surplus}} = \underbrace{p_i+\sigma y}_{\text{match productivity}} - \underbrace{\left[b_I+\theta_i\frac{\beta}{1-\beta}\gamma+\frac{\lambda}{r+\lambda+\theta_i q(\theta_i)}(b_U-b_I)\right]}_{\text{worker outside options}}$$

$$+ \underbrace{\mu\mathbb{E}(S_{e_i})}_{\text{continuation surplus after idiosync. shock}} + \underbrace{\sum_{j\neq i}\Omega_{ij}\max[S_{e_j}(y),0]}_{\text{surplus after agg. shock}}$$

where $\mathbb{E}[S_{e_i}] = \int \max[S_{e_i}(y'),0]\,dG(y')$

Equilibrium Conditions: Job Creation

Job Creation (JC)

$$\underbrace{(r+\mu+\sum_{j\neq i}^{n}\Omega_{ij})\frac{\gamma}{q(\theta_i)}}_{\text{Vacancy cost}} = \underbrace{(1-\beta)\left[p_i+\sigma\bar{y}+\mu\mathbb{E}(S_{e_i})-\left(b_{l_i}+\theta_i\frac{\beta}{1-\beta}\gamma+\frac{\lambda}{r+\lambda+\theta_i q(\theta_i)}(b_U-b_l)\right)\right]}_{\text{Firm share of surplus}} + \underbrace{\sum_{j\neq i}^{n}\Omega_{ij}\left[\frac{\gamma}{q(\theta_j)}\right]}_{\text{Change in vacancy cost from macro shocks}}$$

Interpretation:

- $p_i \uparrow \Rightarrow$ Firm surplus $\uparrow \Rightarrow \theta_i \uparrow$
- Higher expected surplus after idiosyncratic shock incentivizes vacancy creation in booms.
- Anticipation of macroeconomic shocks mitigates job creation dynamics.

Equilibrium Conditions: Job Destruction

Job Destruction (JD)

$$\underbrace{p_i + \sigma R_i}_{\text{Minimum productivity}} = \underbrace{b_{l_i} + \theta_i \frac{\beta}{1-\beta}\gamma + \frac{\lambda}{r+\lambda+\theta_i q(\theta_i)}(b_U - b_l)}_{\text{Worker outside option}} - \underbrace{\mu \mathbb{E}(S_{e_i})}_{\text{Labor hoarding}} - \underbrace{\sum_{j \neq i}^{n} \Omega_{ij} \max\{S_{e_j}(R_i), 0\}}_{\text{Marginal job survival after aggregate shock}}$$

Interpretation:

- For a given θ_i, $p_i \uparrow \Rightarrow$ Total match output $\uparrow \Rightarrow R_i \downarrow$
- Labour Hoarding : Anticipated productivity increase after idiosyncratic shock mitigates job destruction.
- Anticipated aggregate shocks reduce marginal job destruction during recession.

Asymmetries in Job Creation and Destruction

Adjustment Mechanisms

- Job destruction and vacancy posting are **forward-looking jump variables**
- Unemployment is a **stock variable**: adjusts gradually via matching

Response to Aggregate Shocks

- **Recession (negative productivity shock):**
  - Reservation productivity R_i shifts upward
  - Marginal matches become unprofitable $\Rightarrow$ **instant job destruction**
  - Vacancies decline simultaneously
- **Boom (positive productivity shock):**
  - Firms post more vacancies immediately
  - Matches form gradually through the matching function
  - Destruction falls simultaneously
  - Employment increases **slowly**

Unemployment rises rapidly in recessions but declines gradually in booms.

An Illustration : SS Across Macroeconomic States

State (i)	**Res. Prod. (R)**	**Tightness (θ)**	**Total U (%)**	**Insured (%)**	**Uninsured (%)**
Boom (1)	1.88	0.83	7.19	6.52	0.67
Intermediate (2)	1.9	0.62	8.26	7.38	0.88
Recession (3)	1.91	0.44	9.81	8.59	1.22

Table: Steady-State Equilibrium Results by Aggregate State

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